

Unit 5

Unit 5: jQuery Mobile — Practical/Oral Exam Notes

Context

Part of [Web Application Development](#) | Practical + Oral Exam Prep | SPPU Third Year IT

High-Yield Oral Topics

- **data-role attribute** — Core of jQuery Mobile
- **Mobile First** — Design philosophy
- **Multi-page in single HTML** — Most asked practical
- **Transitions** — slide, fade, flip
- **Mobile vs Desktop** — Comparison table

MINI PROJECT ANCHOR: Simple SMS Mobile Website

Mobile student info lookup app using jQuery Mobile — single HTML file.

SECTION 1 — MOBILE FIRST

Exam Definition

Mobile First is a design strategy where you design the **smallest screen version first**, then progressively enhance for larger screens. It ensures the core experience works on mobile before adding desktop enhancements.

Mobile vs Desktop Web

Aspect	Mobile Web	Desktop Web
Screen Size	Small (320–480px typical)	Large (1024px+)
Input	Touch (fingers, fat targets needed)	Mouse (precise clicks)
Network	Slower (3G/4G)	Faster (broadband)
CPU/RAM	Limited	Abundant
Navigation	Hamburger menus, swipe	Full nav bars
Font Size	Larger (16px min)	Flexible
Images	Compressed, lazy-loaded	Higher resolution OK

Mobile First Benefit: Faster load times, forces prioritization of core content, better UX for majority of users (60%+ traffic is mobile).

SECTION 2 — JQUERY MOBILE SETUP

CDN Setup (Must Know for Practical)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <!-- CRITICAL: Viewport meta tag for mobile -->
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>SMS Mobile</title>

  <!-- jQuery Mobile CSS -->
  <link rel="stylesheet"
    href="https://code.jquery.com/mobile/1.4.5/jquery.mobile-1.4.5.min.css">
</head>
<body>

  <!-- Pages go here -->

  <!-- jQuery (MUST load before jQuery Mobile) -->
  <script src="https://code.jquery.com/jquery-1.11.3.min.js"></script>
  <!-- jQuery Mobile JS -->
  <script src="https://code.jquery.com/mobile/1.4.5/jquery.mobile-1.4.5.min.js"></script>
</body>
</html>
```

Critical Order: jQuery CSS → HTML Content → jQuery JS → jQuery Mobile JS

SECTION 3 — COMPLETE SMS MOBILE APP

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>SMS Mobile App</title>
  <link rel="stylesheet"
```

```
href="https://code.jquery.com/mobile/1.4.5/jquery.mobile-1.4.5.min.css">

<style>
  .stat-card {
    background: #3498db;
    color: white;
    padding: 15px;
    border-radius: 8px;
    text-align: center;
    margin: 10px 0;
  }
  .stat-card h2 { font-size: 2em; margin: 5px 0; }
</style>
</head>
<body>

<!-- ===== PAGE 1: HOME ===== -->
<div data-role="page" id="home" data-theme="b">

  <div data-role="header" data-position="fixed">
    <h1><img alt="SMS Mobile logo" data-bbox="78 458 92 472"/> SMS Mobile</h1>
    <a href="#menu" data-icon="grid" class="ui-btn-right">Menu</a>
  </div>

  <div role="main" class="ui-content">
    <!-- Statistics Cards -->
    <div class="stat-card">
      <p>Total Students</p>
      <h2>150</h2>
    </div>

    <!-- Navigation Buttons -->
    <a href="#students" data-role="button" data-icon="user"
      data-transition="slide">View Students</a>

    <a href="#register" data-role="button" data-icon="plus"
      data-transition="slide" data-theme="b">Register Student</a>

    <a href="#about" data-role="button" data-icon="info"
      data-transition="fade">About</a>
  </div>

  <div data-role="footer" data-position="fixed">
    <div data-role="navbar">
```

```
<ul>
  <li><a href="#home" data-icon="home" class="ui-btn-active">Home</a></li>
  <li><a href="#students" data-icon="grid">Students</a></li>
  <li><a href="#register" data-icon="plus">Register</a></li>
</ul>
</div>
</div>
</div>
```

<!-- ===== PAGE 2: STUDENT LIST ===== -->

```
<div data-role="page" id="students">

  <div data-role="header">
    <a href="#home" data-icon="back">Back</a>
    <h1>All Students</h1>
  </div>

  <div role="main" class="ui-content">
    <!-- Search Filter -->
    <form>
      <input type="search" id="studentSearch" placeholder="Search students...">
    </form>

    <!-- Listview - JQM's signature component -->
    <ul data-role="listview" data-filter="true" data-inset="true"
      data-filter-placeholder="Filter students...">

      <li data-role="list-divider">IT Department</li>
      <li>
        <a href="#profile">
          
          <h2>Rahul Sharma</h2>
          <p>Roll No: 101 | IT</p>
          <p class="ui-li-aside"><strong>85%</strong></p>
        </a>
      </li>
      <li>
        <a href="#profile">
          <h2>Priya Patel</h2>
          <p>Roll No: 102 | IT</p>
          <p class="ui-li-aside"><strong>92%</strong></p>
        </a>
      </li>

    </ul>
```

```
<li data-role="list-divider">CS Department</li>
<li>
  <a href="#profile">
    <h2>Amit Kumar</h2>
    <p>Roll No: 201 | CS</p>
    <p class="ui-li-aside"><strong>78%</strong></p>
  </a>
</li>
</ul>
</div>

<div data-role="footer">
  <h4>Total: 3 Students</h4>
</div>
</div>

<!-- ===== PAGE 3: REGISTER ===== -->
<div data-role="page" id="register">

  <div data-role="header">
    <a href="#home" data-icon="back">Back</a>
    <h1>Register Student</h1>
  </div>

  <div role="main" class="ui-content">
    <form id="mobileRegForm">

      <!-- Text Input -->
      <div data-role="fieldcontain">
        <label for="mName">Full Name:</label>
        <input type="text" id="mName" placeholder="Enter full name" required>
      </div>

      <!-- Email Input -->
      <div data-role="fieldcontain">
        <label for="mEmail">Email:</label>
        <input type="email" id="mEmail" placeholder="student@example.com">
      </div>

      <!-- Select Menu -->
      <div data-role="fieldcontain">
        <label for="mCourse">Course:</label>
        <select id="mCourse" data-native-menu="false">
          <option value="IT">Information Technology</option>
```

```
<option value="CS">Computer Science</option>
<option value="ENTC">Electronics</option>
</select>
</div>

<!-- Flip Toggle -->
<div data-role="fieldcontain">
  <label for="mActive">Active Student:</label>
  <select id="mActive" data-role="flipswitch">
    <option value="yes">Yes</option>
    <option value="no">No</option>
  </select>
</div>

<!-- Slider -->
<div data-role="fieldcontain">
  <label for="mYear">Year:</label>
  <input type="range" id="mYear" min="1" max="4" value="3"
    data-highlight="true">
</div>

  <button type="submit" data-role="button" data-icon="check"
    data-theme="b">Register</button>
</form>

<!-- Collapsible section -->
<div data-role="collapsible" data-collapsed="true">
  <h3>Additional Info</h3>
  <p>Fill the form above to register a new student.</p>
</div>
</div>

<div data-role="footer">
  <h4>SMS – Student Portal</h4>
</div>
</div>

<!-- ===== PAGE 4: ABOUT ===== -->
<div data-role="page" id="about" data-theme="a">
  <div data-role="header">
    <a href="#home" data-icon="back">Back</a>
    <h1>About</h1>
  </div>
  <div role="main" class="ui-content">
```



```
<h2>Student Management System</h2>
<p>Mobile web app built with jQuery Mobile.</p>
<p>Version: 1.0 | SPPU Third Year IT</p>
</div>
</div>

<!-- Scripts -->
<script src="https://code.jquery.com/jquery-1.11.3.min.js"></script>
<script src="https://code.jquery.com/mobile/1.4.5/jquery.mobile-1.4.5.min.js"></script>

<script>
$(document).on('pagecreate', '#register', function() {
    $('#mobileRegForm').on('submit', function(e) {
        e.preventDefault();
        const name = $('#mName').val();
        const course = $('#mCourse').val();

        if (!name) {
            alert('Please enter student name');
            return;
        }

        // Show success popup
        $('<div>')
            .text(`${name} registered in ${course}!`)
            .attr('data-role', 'popup')
            .appendTo('#register')
            .popup('open', { positionTo: 'window', transition: 'pop' });

        // Clear form
        this.reset();
        $.mobile.resetActivePageHeight();
    });
});

// pageshow – fires every time page becomes visible
$(document).on('pageshow', '#students', function() {
    console.log('Students page shown – could refresh data here');
});
</script>
</body>
</html>
```

data-role	Element	Purpose
page	<div>	Single mobile screen/page
header	<div>	Top bar with title
footer	<div>	Bottom bar
navbar	<div>	Navigation bar (in header/footer)
listview		Mobile-styled list
list-divider		Section header in list
button	<a> or <button>	Touch-friendly button
collapsible	<div>	Expandable section
fieldcontain	<div>	Form field wrapper
flipswitch	<select>	Toggle switch
popup	<div>	Modal popup

Page Transitions

Transition	Effect
slide	Slides left/right (default)
slideup	Slides upward
slidedown	Slides downward
fade	Fades in/out
flip	3D flip
pop	Pops from center
none	No animation

```
←!— Apply transition to link →
<a href="#page2" data-transition="slide">Go to Page 2</a>
<a href="#page2" data-transition="flip" data-direction="reverse">Go Back</a>
```

JQM Themes

Swatch	Color
data-theme="a"	Dark (black/grey)
data-theme="b"	Blue/Primary


```
<!-- Apply theme to page, button, or listview -->
<div data-role="page" data-theme="b">
  <a href="#" data-role="button" data-theme="b">Primary Button</a>
```

VIVA Q&A — Unit 5

Q1: What is jQuery Mobile?

jQuery Mobile is a touch-optimized, HTML5-based UI framework for building mobile web applications. It works on all mobile platforms (iOS, Android, Windows Phone) and uses `data-role` attributes to automatically enhance HTML elements into mobile-friendly widgets.

Q2: What is `data-role` in jQuery Mobile?

`data-role` is an HTML5 data attribute used by jQuery Mobile to identify elements and enhance them into mobile UI components. For example, `data-role="page"` defines a mobile screen, `data-role="listview"` transforms a `<ul>` into a styled mobile list.

Q3: How does jQuery Mobile handle multi-page navigation?

In a single HTML file, you can have multiple `<div data-role="page">` elements, each with a unique `id`. Clicking `<a href="#page2">` causes JQM to use **AJAX** to smooth-transition to that page ID without reloading the browser. This creates an app-like experience.

Q4: What is Mobile First design?

Mobile First means designing for the smallest screen (mobile) first, then adding enhancements for larger screens. Benefits: Forces content prioritization, faster load on mobile, CSS media queries add features progressively rather than hiding them.

Q5: What are JQM page events? How are they different from `$(document).ready()` ?

JQM uses AJAX to load pages, so `$(document).ready()` fires only once on initial load. JQM page events fire per-page: `pagecreate` (DOM ready, initialize widgets), `pageshow` (page visible — refresh data), `pagehide` (navigating away — cleanup). Use `$(document).on('pageshow', '#myPage', fn)`.

Oral Trap Questions

Trap: "Is jQuery Mobile still relevant?"

jQuery Mobile is older technology (less active development). Modern alternatives include React Native Web, Ionic, and PWAs. However, for simple mobile web projects and academic assignments, JQM is quick to implement and covers all mobile UI needs with minimal code.

Trap: "Can jQuery Mobile work offline?"

Yes, by combining with the HTML5 **Application Cache** or **Service Workers** (PWA). JQM itself doesn't provide offline support, but since it's a web app, the browser can cache files. For full offline support, implement a Service Worker.

"I created a mobile website using jQuery Mobile with 4 pages in a single HTML file: Home, Student List, Register, and About. The Home page has statistics and navigation buttons. The Student List uses `data-role="listview"` with built-in filter. The Register page uses a mobile form with select menus, slider, and flipswitch. Page transitions use `data-transition="slide"`. I used `pagecreate` event for form submission logic."

High Probability Questions

1. What is jQuery Mobile? How is it different from jQuery?
2. Explain `data-role` with 5 examples
3. How do you create multi-page app in jQuery Mobile?
4. What is Mobile First design? Advantages?
5. What are page transitions? Name 3 types
6. What is `listview` in jQuery Mobile?
7. Explain JQM page events (`pagecreate`, `pageshow`, `pagehide`)
8. What is the viewport meta tag? Why is it critical?
9. Mobile Web vs Desktop Web differences
10. What are JQM themes?